Development and Application of the Smart Helmet for Disaster and Safety

Mingi Jeong
IoT Research Division
Electronics and Telecommunications Research Institute (ETRI)
Daejeon, Republic of Korea
mgjeong@etri.re.kr

Hyesun Lee
IoT Research Division
Electronics and Telecommunications Research Institute (ETRI)
Daejeon, Republic of Korea
hyesun.lee@etri.re.kr

Myungnam Bae
IoT Research Division
Electronics and Telecommunications Research Institute (ETRI)
Daejeon, Republic of Korea
mnbae@etri.re.kr

Dong-Beom Shin
IoT Research Division
Electronics and Telecommunications Research Institute (ETRI)
Daejeon, Republic of Korea
sdb@etri.re.kr

Sun-Hwa Lim
IoT Research Division
Electronics and Telecommunications Research Institute (ETRI)
Daejeon, Republic of Korea
limsh@etri.re.kr

Kang Bok Lee
IoT Research Division
Electronics and Telecommunications Research Institute (ETRI)
Daejeon, Republic of Korea
kblee@etri.re.kr

***Abstract*— This paper introduces the development of the smart helmet needed to respond to accidents of rescue workers in the event of a disaster. With the emergence of many IoT-based devices and applications, many of the services that use them are active. However, each service was developed in a specific field, making it difficult to apply new devices, modify application and make changes to services. With the advent of a variety of IoT-based devices and services, we need middleware that can easily integrate them. Our researchers have developed the new software framework enable to integrate a wide range of devices and services and efficiently manage resources. In addition, based on this, we develop the smart helmet to respond the disaster safety accidents. The smart helmet collects, generates, and converts information on sensors (infrared camera, electro optical camera, drone camera, oxygen residual sensor, 6-axis inertial sensor, and smart watch) and it can be monitored through head mounted display(HMD) and the Command Center. Finally, we developed a simulator and generated data based on scenarios, and also tested all devices and service outdoors.**

Keywords*— *smart helmet, disaster response, disaster and safety, software framework, IoT

I. INTRODUCTION

Currently there are many fields where IoT devices are used. Especially, The public sector also needs to use the Internet of Things to respond to disasters [1,2]. Services using IoT devices, mobile and cloud services have also been developed to respond to disasters [3]. However, conventional studies are not general-purpose, device-dependent, and difficult to easily apply or manage service devices and applications because they have been performed for specific purposes only.

Now a lot of IoT devices or application services being developed fast and internationally, a framework that can be easily used, integrated and developed with the devices is needed[4]. In order to achieve this goal, this study developed a software framework and demonstrated service devices and applications to response disaster safety accident using it.

This study introduces the development and application of smart helmet service applied with a variety of devices, such as an infrared camera, an electro optical camera, a drone camera, an inertial sensor, and a smart watch. Although this is a smart helmet for responding to disaster safety accidents, it is easy to apply to other areas by using the software framework that has been developed.

Section 2 introduces the key features of the software framework developed, and Section 3 introduces smart helmet system structure using it. Section 4 introduces the results and overall service of smart helmet devices developed. Section 5 introduces the results of a simulator and outdoor demonstration results based on disaster safety accident response scenarios using smart helmet developed. Section 6 introduces the conclusions of developing smart helmets for responding to disaster safety accidents and future research directions.

II. SOFTWARE FRAMEWORK DEVELOPMENT

A. Outline

Complex services such as disaster safety response service can be developed if it meets the interworking with various devices and efficient resource management system. In order to integrate and operate a wide variety of devices and services, the software framework should provide flexibility, resource management convenience, resource reliability, security, distributed processing functions, big data resource of management and cognitive function[5]. In particular, it was necessary to develop a framework that takes up less processor resource and memory while providing only the core functions of the software framework applicable to the IoT device.

B. Flexibility and Resource Management Convenience

The core framework calls for basic resource management functions using REST(representational state transfer) style API, so resource management is convenient. It is easy to link and integrate with other devices and services, and easily allows the registration and release of devices and services according to the service or environment.

Basic functions of resource management include create, retrieve, update, delete, and notify, which also supports xml method for managing resources without restriction. The example code and schema for the basic functions are shown in figure 1. The example code is subscription creation code that includes subscription name, subscriber Uniform Resource Identifier(URI), and creator URI. It has also notification criteria that can set container instance data

threshold to notify. And, the example schema is about notify message. It basically includes subscription name, target resource name, and depending on the option of the subscription, it selectively includes any of the target resource's status(create, retrieve, update, delete), resource content, and attributes.

```
<operation>Create</operation>
<senderURI>"dios/applications/roomConditioningApp0"</senderURI>
<resourceContents>
    <subscription>
    <subscriptionName>"dios/devices/temperatureSensor0/containers/temperatureContainer/roomConditioningSub0"</subscriptionName>
        <subscriberURIs>"dios/applications/roomConditioningApp0"</subscriberURIs>
        <creatorURI>"dios/applications/roomConditioningApp0"</creatorURI>
        <notificationCriteria>
            <criteria>  //or
                <valueCriteria>  // and
                    <threshold>
                        <name>"t"</name>
                        <valueRange>"(RANGE_ABOVE_UPPER,0,30)"</valueRange>
                    </threshold>
                </valueCriteria>
            </criteria>
            <criteria>  //or
                <valueCriteria>  // and
                    <threshold>
                        <name>"t"</name>
                        <valueRange>"(RANGE_BELOW_LOWER,18,0)"</valueRange>
                    </threshold>
                </valueCriteria>
            </criteria>
        </notificationCriteria>
        <notificationType>MethodCall</notificationType>
        <notificationContentType>TYPE_SUBS_CHILD_RES_STATUS</notificationContentType>
    </subscription>
</resourceContents>
```

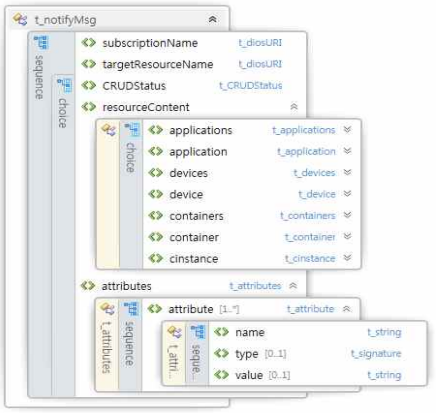


Fig. 1. Subscription create example(up), Notify schema(down),

C. Resource Reliability

The core framework can completely restore resource of the tree structure information that was stored and managed before shutdown if the program was normal/unusual shut down and restarted. Because It uses with both memory and the SQLite engine, and distinguishes that the service device and the application are restarted in a normal state or not. Reliable resources are available because the device or application can be reconstructed in an independent process based on the service device or application information in the restored resource tree.

D. Security

The right to access or control specific devices can be granted to specific device or service users. It is necessary to restrict the use of specific functions of devices while developing various IoT devices and services, and have security that requires only a few users to use. For example, in the case of drone control in smart helmet services developed, If all rescue workers control the drones, they may not be able to perform desired/efficient rescue operations. Thus, the control of the drone was authorized to only the team leader of the rescue workers. figure3 is a picture of a drone camera controlled by a board mounted on a team leader's helmet.

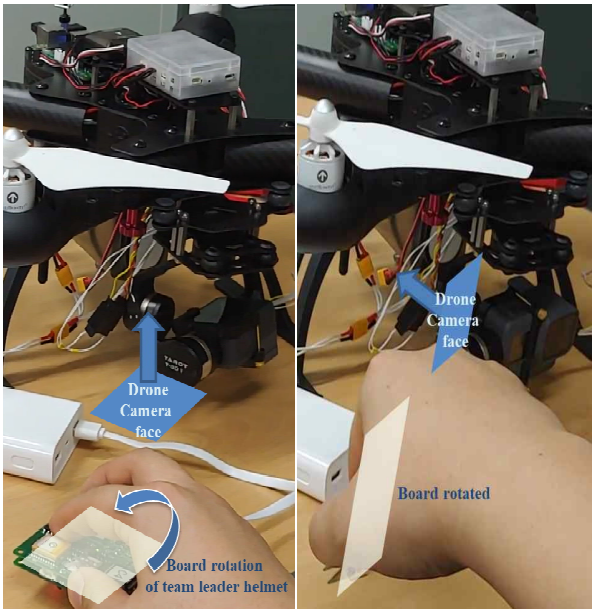


Fig. 2. The gaze of the drone camera controlled by the team leader's helmet-mounted board

By managing the configuration, only the users of the device or service that are authorized by this configuration can use the functions of each device or application.

E. Big Data Resource of Management and Cognitive Function

It provides big data resource management and cognitive function. Information collected from the device is stored and managed in internal memory or database by distributed processing. It provides the ability to process and analyze stored resources, subscribe resources and notify in a variety of method.

1) Resource subscription method

By Setting criteria for stored information, it provides subscription and notification to specific information. If the target information collected and stored from the device meets certain criteria, then a message is given to the receiver the notification. The definition of criterion is divided into four main categories:

- Time criteria: Define the conditions related to the creation/update time of the target resource
- Status criteria: Define the conditions related to the state(create, update, delete) of the target resource
- Attribute criteria: Define the conditions related to the properties of the target resource
- Value criteria: Define the conditions related to the value of the target resource's container instance

In addition, in the definition of a condition, the target resource can be a resource or child resource, and each criterion provides a variety of optional features, such as something greater than or less than a certain value, matching, including, or going out.

2) Resource subscription method

There are two ways of giving notifications: "Method Call" or "Session-less Signal." There is a method call that sends messages to the target subscribers, and its processor being handled receives that messages sequentially. There is also session-less signal called a signal call can notify to every receivers, and these receivers first process for receiving that notification.

Different rule-based events can be generated by setting up and changing notification methods according to a service or situation scenario. For example, a rule-based cognitive function based on the standard operating procedure(SOP) can be added so that rescue workers can analyze and provide information on issues that could arise in the event of a disaster, enabling them to provide efficient rescue and emergency services.

3) Real-time management of resources

Each information is registered as a resource for the framework and provides create, retrieve, update, deletion, subscription and notification functions, which are performed by distributed processing. The distributed resources can be provided by convergence, execution and control. Because of these distributed system, even when big data being collected from various devices, the software framework can provide data collection and criteria-based alert in real time. Using the software framework, the smart helmet service needed to respond to disaster safety accident can be developed. More detailed descriptions of the smart helmet service are given in Section 3.

III. SMART HELMET STRUCTURE USING THE SOFTWARE FRAMEWORK

Smart Helmet for responding to disasters is an IoT-based helmet system technology that provides efficient rescue and emergency services for rescue workers by collecting, generating, converting and sharing the information necessary to respond to an outdoor disaster site.

The smart helmet system consists of Bio & Framework Subsystem(BFS), Multimedia Processing Subsystem (MPS), and Communication Subsystem (CPS) as shown in figure 4.

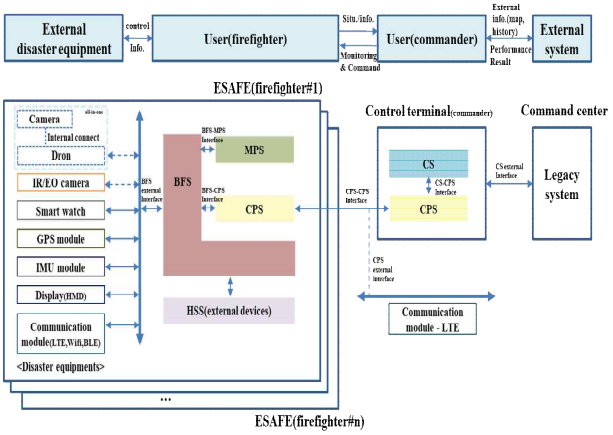


Fig. 3. Smart helmet system structure

BFS is a software framework system that provides information sharing service by linking all devices introduced in Section 4 to other systems. And, it can collect, analyze and process bio information of rescue workers.

MPS is a system that provides thermal infrared cameras, visible light cameras, connection of drone cameras, collection and delivery of multimedia information, and visualization through head mounted display(HMD).

CPS is a subsystem that provides inter-structural communication and communication between rescuers and command posts. The bio and location information of the rescuer is sent to the command post. It is also a system that can send voice and video information of rescuers to the command post and share voices between rescuers.

The command subsystem is a system that operates the services mounted in the command center and monitors the bio, location and image information of all fire crews by sharing them.

In addition, The smart helmet system include Helmet Structure Subsystem (HSS) and external system as shown in figure 4. HSS refers to external devices such as button controllers, which consist of buttons that can provide input from both power and external sources, and light helmet devices that facilitate long-term operation and can incorporate displays and multiple communication devices. Also, an external system refers to a system that can download the map needed to indicate the location of the rescue workers on the map and share operational results.

IV. INTROCUTION OF SMART HELMET FOR DISASTER SAFETY ACCIDENT RESPONSE

The smart helmet and external devices are worn in wearable form and are configured in rescue sites to communicate with each other under the command center mediation and provide services for responding to disaster safety accidents. As shown in figure 5 and 6, smart helmet are built with integrated camera and HMD. There are also external devices that consist of the button controller and the smart watch. The button controller can power on/off and control HMD and the smart watch can collect bio information of rescue workers.

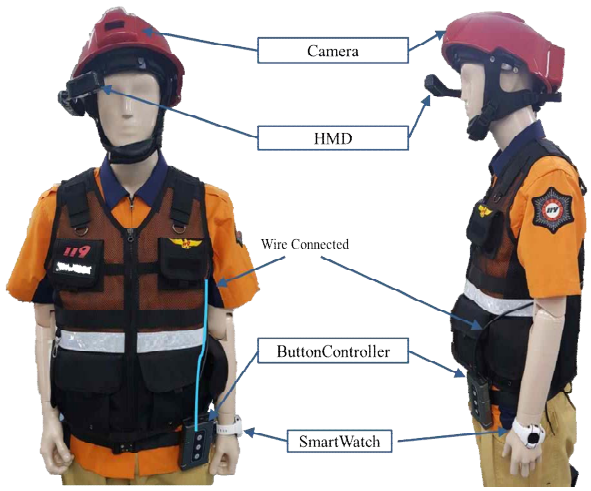


Fig. 4. Wearing smart helmets and external devices

The board in smart helmet is embedded with GPS, IMU, LTE, Wi-Fi, BLE, RDC100(image processing) chips.

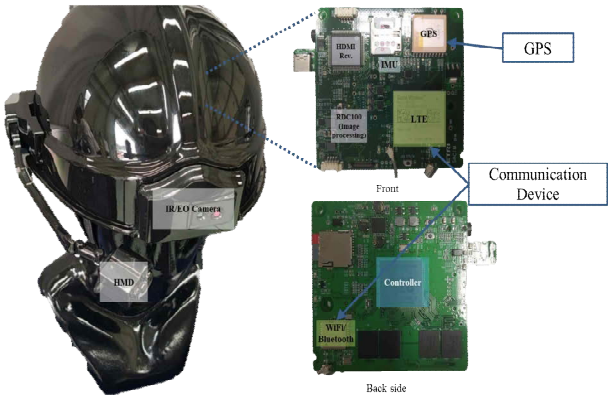


Fig. 5. Smart helmets with integrated cameras and displays

And, the smart helmet services consist of devices and applications that manage bio, media and location information. A detailed picture of this is shown below.

Each service device(e.g. CameraDev, GPSDev, SWDev, DisplayDev) provides the status and control of the device in common, in addition, a smart watch service device collects bio data, and a display service device provides function to select the image displayed on the screen. And each service application(e.g.CameraApp, GPSApp, SWApp, DisplayApp) provides device's status subscription in common, in addition, a smart watch service application can subscript bio data and notify emergency situation, and a display service application can notify when the mode of the display is chaged.

A device controller first select target device to control. Second, if either a camera, a GPS, or a smart watch is selected, the status(start, stop, on, off) of the device can be controlled, and if a display is selected, display mode(IR view, EO view, drone view, text message, display off) can be controlled.

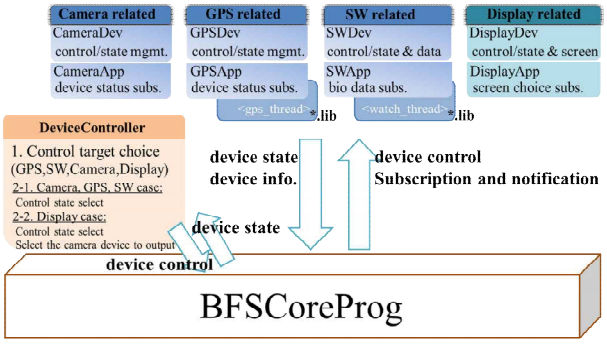


Fig. 6. Service device and application architecture plot

A. Bio Information

Bio information refers to pulse, body temperature and motion information collected from smart watches worn on the wrist by rescue workers.

Smartwatch is registered as a service device to the smart helmet, service application manages status (start, end, run, stop) information of the device, analyzes collected bio information and alerts in case of abnormal situations. Abnormalities are indicated by the rescue worker's pulse that is outside a certain standard, or when the body temperature becomes hot or cold outside a certain standard, and when motion is absent or resumed, the worker alarm is indicated. Command control monitors the bio information of each team's rescue workers through text & graphs in real time and alerts them of abnormal situations shown in figure 7.

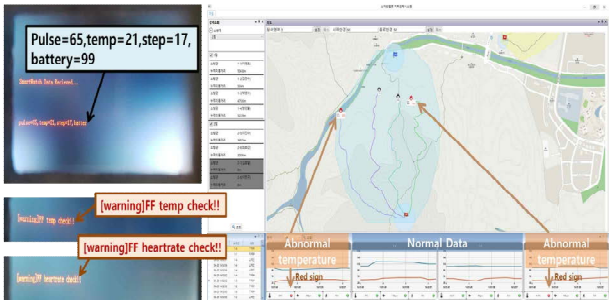


Fig. 7. Bioinformation and notifications on HMD(left), Command Control Screen (right)

B. Media Information

Smart Helmet has developed a service device for IR(infrared) cameras, EO(electro optical) cameras, drone cameras, and HMD, and the service application can manage the status of each device and select the screen to display: drone, IR or EO. through buttons connected to helmets. It also transmits the images of each camera to the command center. Command center can monitor the images of all rescue workers and drone simultaneously as shown in figure 9.

Because drones can monitor nearby accident sites in case of accident response such as rescue situation and extinguishing the fire, proving eye matching function between drone and team leader, firefighters can control drones at the same time as responding to accidents. The eye matching function rotates the drone's camera in accordance with head direction of a rescue worker's leader in the three axis rotation of the board mounted on the helmet. And, it also be integrated with the HMD. The drone hovers the accident scene, the camera mounted on the drone is controlled in line with the view of the previously authorized rescue team leader, and the scene of the drone camera is shared with all rescue workers and command center.

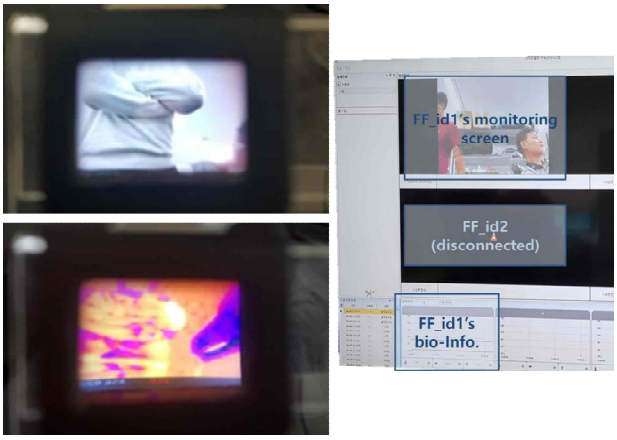


Fig. 8. HMD for EO/IR camera image(left), commander screen (right)

C. Location Information

The smart helmet contains GPS, develop and manage GPS service devices/applications. In addition to the status of the GPS device, the number of satellites connected to provide reliable information is analyzed so that if more than a certain number of satellites are satisfied, the location of rescuer can be transferred to the command center shown in figure 9. Also, this service provides notification in three situations:

- Duplicating zone where other rescuers have searched
- Leaving the field of search,
- Leaving alone the location of other rescuers.

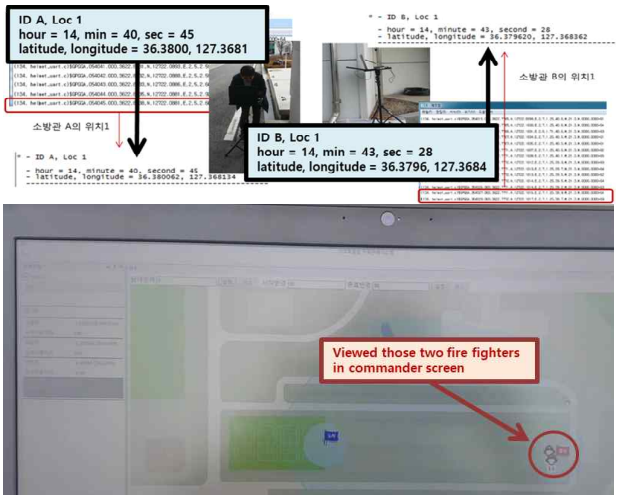


Fig. 9. Firefighter's GPS information view in terminal(up), acquisition of location for firefighter 1 and 2 in commande center laptop screen (down)

D. Communication

The internal devices and the smart helmet communicate IPC. The smart helmet and devices use TCP, UDP socket communications shown in figure 11. And, fire fighters(FFs) share drone's camera image in Wi-Fi. The smart helmet and command center exchange the information to each other in LTE.

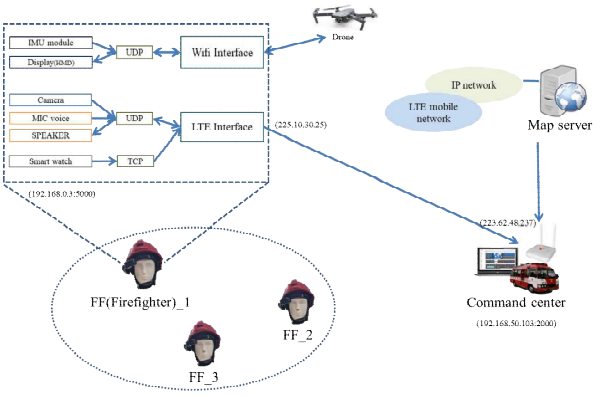


Fig. 10. Network Structure between Smarthelmet, Drones, and Moving Control

V. Disaster Safety Accident Response Scenarios-Simulator and Outdoor Test Results

The initial sequence is performed when the smart helmet is powered on and started. When power is on, each service device and application is executed in order and indicate that there is any error or not. They also register subscriptions to receive notifications when they meet certain criteria for information collected in real time.

GPS information collection is carried out by the sequence shown in Figure 13. First, GPSDev collects GPS information. Second, If the number of satellites meets more than 5, GPSApp sends the data to the command center by judging that reliable GPS information is collected. And, the GPS information interface consists of rescuer id, time information, latitude, longitude, and altitude.

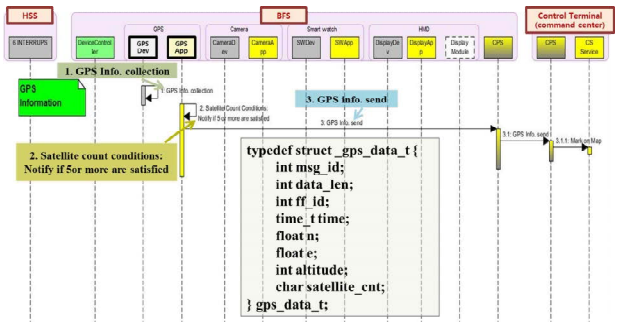


Fig. 11. GPS Information Collection Sequence & GPS interface

Bio/motion information collection is performed by the sequence shown in the figure 14, and the bio interface consists of rescuer member id, time information time, body temperature, pulse. The interface of movement information consists of the rescuer id, time information time, and motion as shown in the figure 14.

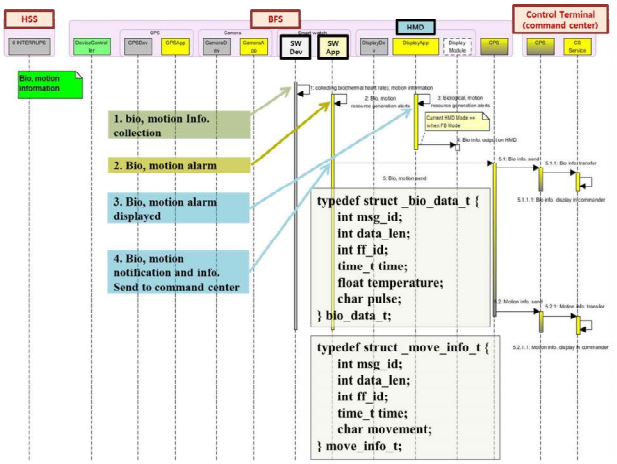


Fig. 12. Bio/motion Information Collection Sequence & bio/motion interface

Other procedures include notification sequence of abnormal bio and motion, switching HMD mode sequence, sending information to the command center sequence, and finally shutting down and restarting sequence.

A scenario has been developed to test each function and sequence. The location data was generated using the QGIS

ver.3.0 tool to accommodate the scenario, and bio and motion data was generated using the Excel. Finally, a scenario where each function is combined consists of six members and two teams of rescuers of Team 1 has rescuer (called ff) id 1, 2, 3, 4 and Team 2 has 5, 6 as shown in figure 12. Each rescue worker is received temperature, pulse, motion and GPS information. And, rescuer is notified for bio, move warning during accident response, and duplicating zone where other rescuers have searched, leaving the scope of their search, and leaving alone the location of other rescuers.

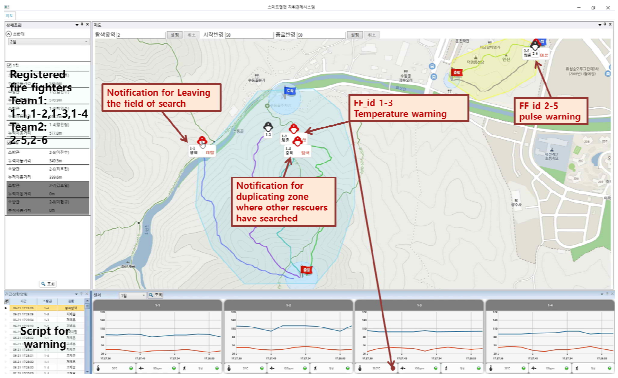


Fig. 13. Commander screen - result of scenario simulator

Two test workers were required to walk around wearing equipment, and the command center received messages from workers such as bio, location and abnormal information shown in figure 16. As a result, it was printed on the monitor for normal bio and location information of workers, and it was found that the two workers received an abnormal notification about duplicating areas of their search shown in figure 17.



Fig. 14. Preparation for an outdoor experiment - PC for outdoor experimental monitoring and debugging(left), PC for commander screen(right)



Fig. 15. Outdoor experimental picture

VI. Conclusion

In this paper, we developed a software framework that allows diverse IoT-based devices and services to be easily integrated and efficiently managed. Using this framework, a smart helmet was integrated with sensors (thermal camera, visible light camera, drone camera, oxygen remaining sensor, inertia sensor, smartwatch), HMD and command center system. In addition, Based on the scenario, we tested the smart helmet with a simulator, and we've completed the actual field trials. we were able to subscribe and notify the bio information of the rescuers in real time, and we were able to identify the location-based search and notification functions for the search area between the rescuers. In the future, the rule-based the Drools tool will be integrated with the Software framework. It will converge information on sensors linked to smart helmets and develop a framework that can support the decision making of rescue workers based on the standard operating procedure(SOP) for disaster site. In addition, the framework developed is a smart helmet not only for disaster response, but it is expected to be applicable in many other areas.

Acknowledgment

This research was supposed by the Field-oriented Support of Fire Fighting Technology Research and Development Program funded by the National Fire Agency (MPSS-Fire Safety-2015-83).

References

[1] Zahra, S. A., *et al.*, "Internet of Things (IoTs) for Disaster Management," Sukkur IBA Journal of Computing and Mathematical Sciences, vol.2, no.1, pp. 77-85, 2018,

[2] Gaire, R., *et al.*, "Internet of Things (IoT) and Cloud Computing Enabled Disaster Management," arXiv preprint arXiv:1806.07530, 2018.

[3] Chung, K., Park, R.C., "P2P cloud network services for iot based disaster situations information. Peer-to-Peer Netw. Application," vol. 9, no. 3, pp. 566–577, 2016.

[4] Hossain, A., *et al.*, "A smartphone-assisted post-disaster victim localization method," IEEE 18th International Conference on High Performance Computing and Communications, pp. 1173–1179. IEEE 2016.

[5] Hyesun, L., *et al.,* "An Internet of Things System Architecture for Aiding Firefighters in the Scene of Disaster," World Congress on Information Technology Applications and Services 2018